

# Trinity 2007. - Half yearly

## Question 1 (Start a new page)

12 marks

(a) Simplify :  $\frac{2}{3} - \frac{x-1}{2}$  2

(b) Express  $\frac{1}{1-2\sqrt{3}}$  in the form  $a+b\sqrt{3}$ , where  $a$  and  $b$  are rational numbers. 2

(c) Find the values of  $x$  that satisfy the inequality  $9-2x < 17$  2

(d) Integration  
Find a primitive function of  $6x - x^{\frac{1}{2}}$  2

(e) If  $\tan x = \frac{2}{3}$  find the exact value of  $\cos x$  if  $x$  is acute. 2

(f) Solve  $2^{3x} = 3$  correct to 2 decimal places. 2

Question 2 (start a new page)

12 marks

- (a) Let  $\alpha$  and  $\beta$  be the roots of the equation  
 $x^2 - 5x + 2 = 0$

Find the values of the following:

- i)  $\alpha + \beta$  1
- ii)  $\alpha\beta$  1
- iii)  $(\alpha + 1)(\beta + 1)$  2

- (b) For what values of  $m$  will the equation  $x^2 - mx + (8 + m) = 0$   
Have:

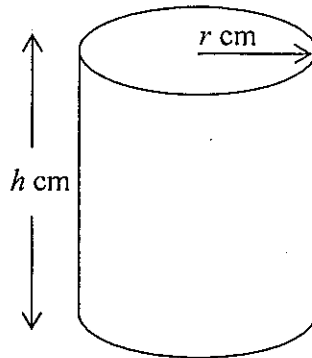
- i) Real and unequal roots? 2
- ii) Roots which are reciprocals of each other? 2

- (c) A parabola has equation  $(x - 2)^2 = -16(y - 3)$

- i) Find the coordinates of its vertex 1
- ii) Find the coordinates of its focus 1
- iii) Sketch the parabola clearly showing the position  
and equation of its directrix. 2

**Question 3 (start a new page)****12 marks**

- (a) A can of baked beans is in the shape of a closed cylinder with height  $h$  cm and radius  $r$  cm, as shown in the diagram below. The volume of the can is  $500\text{cm}^3$ .



- i) Show that the surface area,  $S\text{cm}^2$ , of the can is given by

**2**

$$S = 2\pi r^2 + \frac{1000}{r}$$

- ii) If the surface area,  $S$ , of metal used to make the can is to be minimised, find the radius,  $r$ , of the can.

**3**

- (b) For the function  $f(x) = x^3 - 3x^2 - 9x + 15$ ,

- i) Show that  $\frac{dy}{dx} = 3(x-3)(x+1)$

**1**

- ii) Find the coordinates of any stationary points and determine their nature.

**2**

- iii) Find the coordinates of any point(s) of inflexion.

**2**

- iv) Sketch the curve  $y = f(x)$  showing all important features.

**2**

**Question 4 (start a new page)**

**12 marks**

(a) Differentiate the following with respect to  $x$ :

i)  $\frac{1}{x^4}$

**2**

ii)  $(3x^2 + 4)^5$

**2**

iii)  $\frac{4-x}{1+x^2}$

**2**

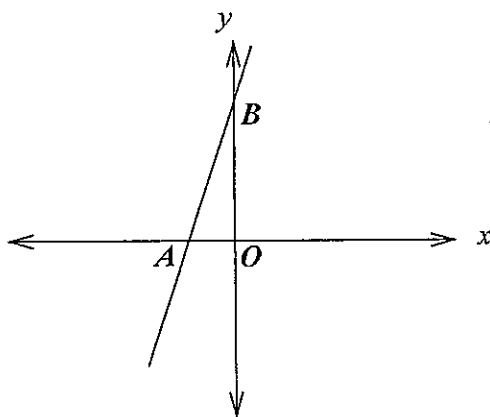
(b) i) Show that the equation of the tangent to the curve  $y = 4 - x^2$  at  $x = -1$  has equation  $2x - y + 5 = 0$

**2**

ii) The tangent  $2x - y + 5 = 0$  cuts the  $x$  and  $y$  axis at  $A$  and  $B$  respectively.

**2**

Calculate the exact area of  $\triangle AOB$  where  $O$  is the origin.



(c) Find the equation of the curve  $g(x)$ , given the curve passes through the point  $(-1, 2)$  and the gradient function is given by  $g'(x) = 3x^2 - 2x + 1$ .

**2**

Question 5 (start a new page)

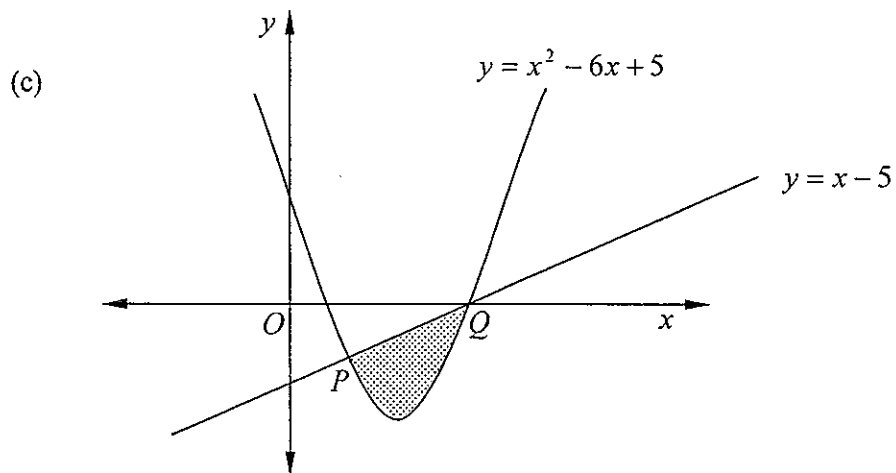
12 marks

(a) Find  $\int (1 + 2x)^3 dx$

1

(b) Evaluate  $\int_1^3 \frac{x^3 + 2x^2 + x}{x} dx$

2



The diagram shows the graphs of the functions  $y = x^2 - 6x + 5$  and  $y = x - 5$  where  $P$  and  $Q$  are the points of intersection.

- i) Find the  $x$  values of  $P$  and  $Q$ .

2

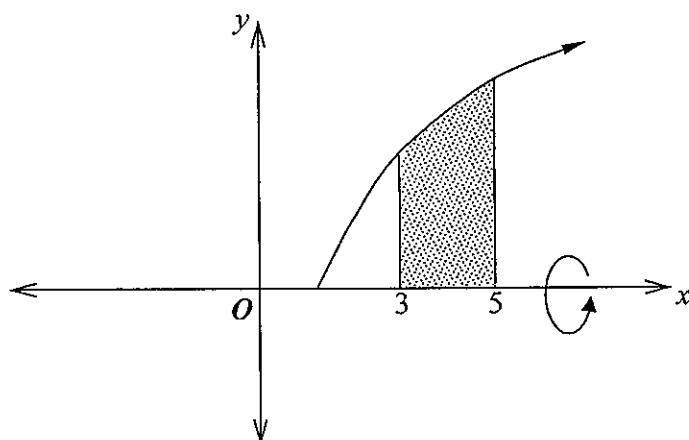
- ii) Calculate the exact area of the shaded region.

2

Question 5 (continued)

- (d) Find the volume of the solid generated when the area between the curve 2

$y = \sqrt{x^2 - 3}$  and the  $x$ -axis bounded by the ordinates  $x = 3$  and  $x = 5$   
is rotated about the  $x$ -axis. Write your answer in terms of  $\pi$ .



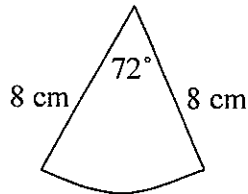
- (e) Use the trapezoidal Rule with 3 subintervals to find an approximation for: 3

$$\int_0^3 x(4-x)^2 dx$$

Question 6 (start a new page)

12 marks

- (a) The figure shows a sector of a circle with radius 8 cm. Find the area of the sector to correct 2 decimal places 2



- (b) Differentiate with respect to  $x$ :

i)  $x \sin 2x$  2

ii)  $4 \tan \frac{x}{3}$  2

(c) Find  $\int (1 + \sin 2x) dx$  2

(d) Find the exact value of  $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \sec^2 2x \, dx$  2

(e) Find the equation of the normal to the curve  $y = 3 \sin 4x$  2  
at the point  $\left(\frac{\pi}{2}, 0\right)$

Question 7 (start a new page)

12 marks

(a) Simplify the following:

2

$$\log_9 25 - \log_3 5$$

(b) Differentiate the following:

i)  $\frac{(\log 2x)^3}{\phantom{00}}$

2

ii)  $xe^{2x} + x$

2

(c) Evaluate  $\int \frac{x^3 - 1}{x^4 - 4x} dx$

2

(d) i) Sketch the graph  $y = e^{2x}$  for  $-2 \leq x \leq 2$

2

ii) Calculate the area under the curve  $y = e^{2x}$   
for  $-2 \leq x \leq 2$

2

Question one.

a)  $\frac{2}{3} - (x-1)$

$= 4 - 3(x-1)$

$= 7 - 3x$

b)  $\frac{1}{1+2\sqrt{3}} \times \frac{1+2\sqrt{3}}{1+2\sqrt{3}}$

$= \frac{1-2\sqrt{3}}{1+2\sqrt{3}}$

$= \frac{1-12}{1+2\sqrt{3}}$

$= \frac{-11}{1+2\sqrt{3}}$

$\therefore a = -1, b = \frac{2}{-11}$

c)  $9 - 2x < 17$

$-2x < 8$

$x > -4$

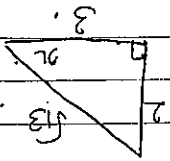
d)  $\int 6x - x^{\frac{2}{3}} dx$

$= 6x^2 - x^{\frac{5}{2}} + C$

$= 8x^2 - 2x^{\frac{1}{2}} + C$

e)  $\tan x = \frac{3}{2}$

$\cos x = \frac{2}{\sqrt{13}}$



$y^2 = 2^2 + 3^2$   
 $y^2 = 4 + 9$   
 $y = \sqrt{13}$

1

1

1

1

Question two.

a)  $x^2 - 5x + 2 = 0$

i)  $\alpha + \beta = -\frac{b}{a}$

$= 5$

ii)  $\alpha\beta = \frac{c}{a}$

$= 2$

iii)  $(\alpha+1)(\beta+1)$

$= \alpha\beta + \alpha + \beta + 1$

$= 5 + 2 + 1$

$= 8$

b)  $x^2 - mx + (8+m) = 0$

i) For real and unequal roots

$\Delta > 0$

$\Delta = b^2 - 4ac$

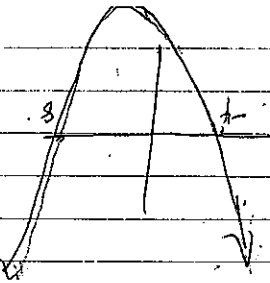
$= m^2 - 4(1)(8+m)$

$= m^2 - 32 - 4m$

$m^2 - 4m - 32 > 0$

$(m-8)(m+4) > 0$

$m < -4, m > 8$



f)

$2^{3x} = 3$

$\log 2^{3x} = \log 3$

$3x \log 2 = \log 3$

$3x = \frac{\log 3}{\log 2}$

$x = 0.53$

1

$\frac{\log 3}{\log 2} = 3x$

$\log_2 3 = 3x$

1

$0.53 = x$

or

ii) let the roots be  $\alpha$  and  $\frac{1}{\alpha}$ .

$$\therefore \alpha \times \frac{1}{\alpha} = \frac{c}{a}$$

$$1 = \frac{8+m}{1}$$

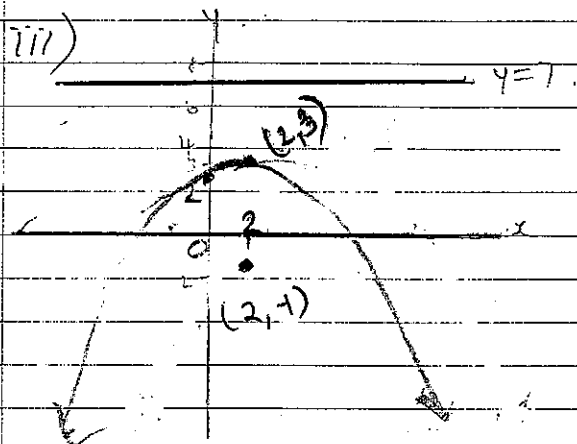
$$m = -7$$

c)  $(x-2)^2 = -16(y-3)$

i) vertex  $(2, 3)$

ii) Focal length  $4a = 16$   
 $a = 4$

$\therefore$  Focus has co-ordinates;  
 $(2, -1)$



### Question Three

a)

Volume of cylinder given by:

$$V = \pi r^2 h$$

$$500 = \pi r^2 h$$

$$h = \frac{500}{\pi r^2}$$

$$\therefore SA = 2\pi r^2 + 2\pi r h$$

$$= 2\pi r^2 + 2\pi r \left( \frac{500}{\pi r^2} \right)$$

$$= 2\pi r^2 + \frac{1000}{r}$$

ii)  $SA = 2\pi r^2 + \frac{1000}{r}$

$$SA' = 4\pi r - \frac{1000}{r^2}$$

For minima  $SA' = 0$

$$4\pi r - \frac{1000}{r^2} = 0$$

$$4\pi r^3 - 1000 = 0$$

$$4\pi r^3 = 1000$$

$$r^3 = 79.5774 \dots$$

$$r = 4.3$$

$$SA'' = 4\pi + \frac{2000}{r^3}$$

$$\text{at } r = 4.3$$

$$= 87.72$$

$> 0 \therefore$  concave up

$\therefore r = 4.3$  gives a minimum Surface Area

b)

$$f(x) = x^3 - 3x^2 - 9x + 15$$

$$\begin{aligned} i) f'(x) &= 3x^2 - 6x - 9 \\ &= 3(x^2 - 2x - 3) \\ &= 3(x-3)(x+1) \end{aligned}$$

ii) stationary points occur when

$$\begin{aligned} f'(x) &= 0 \\ 3(x-3)(x+1) &= 0 \end{aligned}$$

$$x = 3, -1$$

$$\therefore (3, -12) \quad (-1, 20)$$

check  $x = 3$

| x     | 2   | 3 | 4   |
|-------|-----|---|-----|
| f'(x) | -ve | 0 | +ve |

$\therefore x = 3$  is a minimum

check  $x = -1$

| x     | -2  | -1 | 0   |
|-------|-----|----|-----|
| f'(x) | +ve | 0  | -ve |

$\therefore x = -1$  is a maximum

iii) inflexion points occur when  $f''(x) = 0$

$$f''(x) = 6x - 6$$

$$6x - 6 = 0$$

$$6x = 6$$

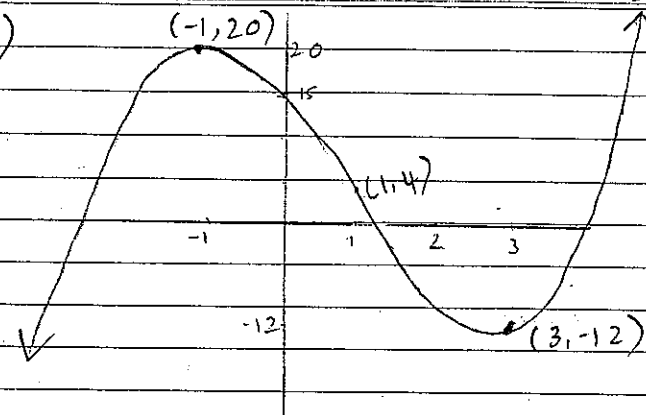
$$x = 1$$

$$(1, 4)$$

| x      | 0   | 1 | 2   |
|--------|-----|---|-----|
| f''(x) | -ve | 0 | +ve |

Change of concavity  
 $\therefore (1, 4)$  is a point of inflexion

iv)



2

### Question four

a)

i) let  $y = \frac{1}{x^4}$

$$\begin{aligned} y &= x^{-4} \\ \frac{dy}{dx} &= -4x^{-5} \\ &= \frac{-4}{x^5} \end{aligned}$$

ii) let  $y = (3x^2 + 4)^5$

$$\begin{aligned} \frac{dy}{dx} &= 5(6x)(3x^2 + 4)^4 \\ &= 30x(3x^2 + 4)^4 \end{aligned}$$

iii) let  $y = \frac{4-x}{1+x^2}$

|             |           |
|-------------|-----------|
| $u = 4-x$   | $u' = -1$ |
| $v = 1+x^2$ | $v' = 2x$ |

$$\begin{aligned} \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{-(1+x^2) - 2x(4-x)}{(1+x^2)^2} \\ &= \frac{x^2 - 8x - 1}{(1+x^2)^2} \end{aligned}$$

b)

$$i) y = 4 - x^2 \quad \text{when } x = -1$$

$$\frac{dy}{dx} = -2x \quad y = 3$$

$$\text{at } x = -1$$

$$= 2$$

$$\text{gradient at } x = -1 \text{ is } 2.$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = 2(x + 1)$$

$$y - 3 = 2x + 2$$

$$2x - y + 5 = 0$$

$$ii) \text{Area} = \frac{1}{2}bh$$

Find co-ordinates A and B.

$$\text{let } x = 0.$$

$$-y + 5 = 0 \quad B.$$

$$y = 5.$$

$$\text{let } y = 0.$$

$$2x + 5 = 0$$

$$x = \frac{-5}{2} \quad A.$$

$$\text{Area} = \frac{1}{2} \times \frac{5}{2} \times 5$$

$$= \frac{25}{4} \text{ units}^2$$

2

c)

$$g'(x) = 3x^2 - 2x + 1$$

$$\int 3x^2 - 2x + 1 \, dx$$

$$= \frac{3x^3}{3} - \frac{2x^2}{2} + x + C$$

$$g(x) = x^3 - x^2 + x + C$$

$$g(-1) = 2$$

$$2 = (-1)^3 - (-1)^2 + (-1) + C$$

$$C = 5$$

$$\therefore g(x) = x^3 - x^2 + x + 5$$

### Question Five

$$a) \int (1+2x)^3 \, dx$$

$$= \frac{(1+2x)^4}{4 \times 2} + C$$

$$= \frac{(1+2x)^4}{8} + C$$

$$b) \int_1^3 \frac{x^3 + 2x^2 + x}{x} \, dx$$

$$= \int_1^3 (x^2 + 2x + 1) \, dx$$

$$= \left[ \frac{x^3}{3} + x^2 + x \right]_1^3$$

$$= \left[ \frac{3^3}{3} + 3^2 + 3 \right] - \left[ \frac{1}{3} + 1 + 1 \right]$$

$$= 18 \frac{2}{3}$$

c)

i) Solve simultaneously.

$$x^2 - 6x + 5 = x - 5$$

$$x^2 - 7x + 10 = 0$$

$$(x-5)(x-2) = 0$$

$$x = 5, 2.$$

$$ii) \int_2^5 (x-5) - (x^2 - 6x + 5) dx$$

$$= \int_2^5 -x^2 + 7x - 10 dx$$

$$= \left[ \frac{-x^3}{3} + \frac{7x^2}{2} - 10x \right]_2^5$$

$$= \left[ \frac{-5^3}{3} + \frac{7(5)^2}{2} - 10(5) \right] - \left[ \frac{-2^3}{3} + \frac{7(2)^2}{2} - 10(2) \right]$$

$$= -4 \frac{1}{6} - (-8 \frac{2}{3})$$

$$= 4 \frac{1}{2} \text{ units}^2$$

d)

$$V = \pi \int y^2 dx$$

$$y = \sqrt{x^2 - 3}$$

$$y^2 = (\sqrt{x^2 - 3})^2$$

$$= x^2 - 3$$

$$V = \pi \int_3^5 x^2 - 3 dx$$

$$= \pi \int_3^5 \frac{x^3}{3} - 3x dx$$

$$= \pi \left[ \frac{5^3}{3} - 15 \right] - \left[ \frac{3^3}{3} - 9 \right]$$

$$= \frac{80\pi}{3} \text{ units}^3$$

e)

$$\begin{array}{ccccc} 2 & 0 & 1 & 2 & 3 \\ y & 0 & 9 & 8 & 3 \\ w & 1 & 2 & 2 & 1 \\ wy & 0 & 18 & 16 & 3 \end{array}$$

$$\begin{array}{ccccc} 2 & 0 & 1 & 2 & 3 \\ y & 0 & 9 & 8 & 3 \\ w & 1 & 2 & 2 & 1 \\ wy & 0 & 18 & 16 & 3 \end{array}$$

$$\begin{array}{ccccc} 2 & 0 & 1 & 2 & 3 \\ y & 0 & 9 & 8 & 3 \\ w & 1 & 2 & 2 & 1 \\ wy & 0 & 18 & 16 & 3 \end{array}$$

$$\sum wy = 37$$

$$\sum wy \times \frac{h}{2}$$

$$h = \frac{1}{2}$$

$$37 \times \frac{1}{2}$$

$$= 18 \frac{1}{2}$$

## Question Six

a) change  $72^\circ$  into radians.

$$\frac{72 \times \pi}{180} = \frac{2\pi}{5}$$

$$\therefore \text{Area} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 8^2 \times \frac{2\pi}{5}$$

$$= 40.21 \text{ cm}^2$$

b)

i) let  $y = x \sin 2x$

$$\frac{dy}{dx} = vu' + uv'$$

$$= \sin 2x + 2x \cos 2x$$

ii) let  $y = \frac{4 \tan x}{3}$

$$\frac{dy}{dx} = \frac{4}{3} \sec^2 \frac{x}{3}$$

c)  $\int (1 + \sin 2x) dx$

$$= x - \frac{1}{2} \cos 2x + C$$

d)  $\int_{\pi/8}^{\pi/6} \sec^2 2x dx$

$$= \left[ \frac{1}{2} \tan 2x \right]_{\pi/8}^{\pi/6}$$

$$= \left[ \frac{1}{2} \tan \frac{\pi}{3} \right] - \left[ \frac{1}{2} \tan \frac{\pi}{4} \right]$$

$$= \left[ \frac{1}{2} \times \sqrt{3} \right] - \left[ \frac{1}{2} \times 1 \right]$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{2}$$

$$= \frac{\sqrt{3} - 1}{2}$$

e)  $y = 3 \sin 4x$

$$\frac{dy}{dx} = 12 \cos 4x$$

at  $x = \frac{\pi}{2}$

$$= 12 \cos 2\pi$$

$$= 12(1) = 12$$

12 is the gradient of the tangent to the curve.

The gradient of the normal is  $-\frac{1}{12}$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{1}{12} (x - \frac{\pi}{2})$$

$$12(y) = -x + \frac{\pi}{2}$$

$$x + 12y - \frac{\pi}{2} = 0$$

$$2x + 24y - \pi = 0$$

## Question seven

a)  $\log_9 25 - \log_3 5 = \frac{\log 25}{\log 9} - \frac{\log 5}{\log 3}$  1

$$= \frac{2 \log 5}{2 \log 3} - \frac{\log 5}{\log 3}$$

$$= 0.$$

b) i) let  $y = (\log 2x)^3$ .

$$\frac{dy}{dx} = 3x \left( \frac{1}{2x} \right) (\log 2x)^2$$

$$= \frac{3}{x} (\log 2x)^2.$$

ii) let  $y = x e^{2x} + 2x$ .

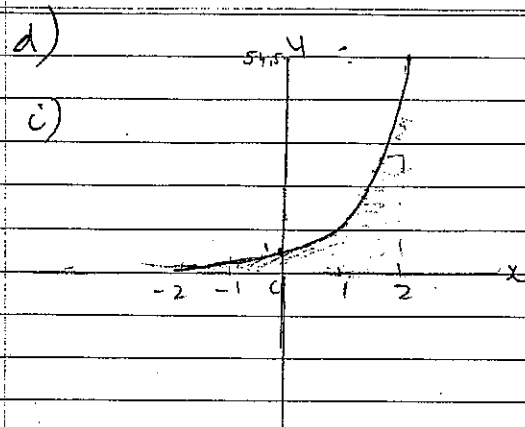
$$\frac{dy}{dx} = e^{2x} + 2x e^{2x} + 2$$

$$= e^{2x} (1 + 2x) + 2$$

c)  $\int \frac{x^3 - 1}{x^4 - 4x} dx$

$$= \frac{1}{4} \int \frac{4(x^3 - 1)}{x^4 - 4x} dx$$

$$= \frac{1}{4} \log_e (x^4 - 4x) + C$$
 2



ii)  $\int_{-2}^2 e^{2x} dx$

$$= \left[ \frac{1}{2} e^{2x} \right]_{-2}^2$$

$$= \left[ \frac{1}{2} e^4 \right] - \left[ \frac{1}{2} e^{-4} \right]$$

$$= \frac{1}{2} (e^4 - e^{-4})$$